Informed Machine Learning Notebook

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1 Overview about these five papers

These five papers show me the overview about informed machine learning:

- taxonomy-level Informed ML,
- logic-constrained neural learning,
- granular computing for confidence-aware outputs.

2 A Unified Optimization Template

A compact mathematical template that covers most methods in this corpus is:

$$\min_{\theta} \; \mathcal{L}_{\text{task}}(\theta) + \lambda_K \, \mathcal{L}_{\text{knowledge}}(\theta) + \lambda_G \, \mathcal{L}_{\text{granularity}}(\theta). \tag{1}$$

Interpretation:

- $\mathcal{L}_{\text{task}}$: standard supervised objective (e.g., cross-entropy, MSE).
- $\mathcal{L}_{\text{knowledge}}$: prior-knowledge term (logic rules, equations, graph constraints).
- $\mathcal{L}_{\text{granularity}}$: uncertainty/credibility term that promotes informative non-point outputs.

Equation (1) is the mathematical counterpart of the taxonomy path "knowledge source → representation → integration location" in the TKDE survey.

3 Logic-Constrained Learning: Three Mathematical Mechanisms

3.1 Hu et al. (logic-net): Posterior Regularization + Distillation

Let $p_\theta(y \mid x)$ be the student network. A rule-regularized teacher $q(y \mid x)$ is constructed by projection:

$$q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}\big(q(Y \mid X) \,\|\, p_\theta(Y \mid X)\big) + C \sum_{\ell} \xi_\ell, \tag{2}$$

subject to rule satisfaction constraints (soft form) using slack variables ξ_ℓ.

A common closed-form structure is:

$$q^*(y \mid x) \propto p_\theta(y \mid x) \exp\left\{ -\sum_{\ell} \gamma_\ell \left(1 - r_\ell(x, y)\right) \right\}, \tag{3}$$

where $r_\ell(x, y) \in [0, 1]$ is soft truth of rule ℓ.

Then student update mixes data labels and teacher supervision:

$$\theta \leftarrow \arg\min_{\theta} \frac{1}{N} \sum_{n=1}^{N} \Big[(1 - \pi_t) \, \ell\big(y_n, p_\theta(\cdot \mid x_n)\big) + \pi_t \, \ell\big(q^*(\cdot \mid x_n), p_\theta(\cdot \mid x_n)\big) \Big]. \tag{4}$$

This is the "teacher-student + rule projection" logic in mathematical form.

3.2 Xu et al. (semantic loss): Constraint-Exact Differentiable Loss

For Boolean outputs $X_1, \ldots, X_n$ with network probabilities $p = (p_1, \ldots, p_n)$ and a logical constraint α, semantic loss is:

$$\mathcal{L}_s(\alpha, p) = -\log \sum_{x \models \alpha} \prod_{i:x_i=1} p_i \prod_{i:x_i=0} (1 - p_i). \tag{5}$$

This directly measures probability mass assigned to satisfying worlds of α. Training usually uses:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \beta\, \mathcal{L}_s(\alpha, p). \tag{6}$$

Key property: logical meaning is preserved exactly at the constraint level, while optimization remains gradient-based.

3.3 Fischer et al. (DL2): Declarative Constraints to Loss and Queries

DL2 maps constraints φ into nonnegative losses $\mathcal{L}(\varphi)$ with:

$$\mathcal{L}(\varphi) = 0 \iff \varphi \text{ is satisfied}, \qquad \mathcal{L}(\varphi) \geq 0, \tag{7}$$

and differentiability almost everywhere.

Representative atom relaxations (hinge-style) are:

$$\mathcal{L}(a \leq b) = \max(0, a - b), \qquad \mathcal{L}(a = b) = |a - b|. \tag{8}$$

Complex formulas are recursively composed.

Training with constraints can be written as:

$$\min_{\theta}\ \mathcal{L}_{\text{task}}(\theta) + \lambda\, \mathbb{E}_x\big[\mathcal{L}(\varphi(x, \theta))\big]. \tag{9}$$

Querying is posed as input optimization:

$$x^* = \arg\min_{x}\ \mathcal{L}(\psi(x, \theta)), \tag{10}$$

where ψ encodes the query constraints.

4 From Point Predictions to Granular Predictions

4.1 Takagi–Sugeno Base Model (numeric)

For rule i:

$$R_i:\ \text{if } x_1 \text{ is } A_{i1}, \ldots, x_m \text{ is } A_{im},\ \text{then } y = b_i, \tag{11}$$

with firing strength

$$w_i(x) = \prod_{j=1}^{m} \mu_{A_{ij}}(x_j). \tag{12}$$

Output:

$$\hat{y}(x) = \frac{\sum_{i=1}^{c} w_i(x) b_i}{\sum_{i=1}^{c} w_i(x)}, \quad \mathcal{L}_{\text{task}} = \sum_k (\hat{y}(x_k) - y_k)^2. \tag{13}$$

4.2 Granular Elevation

Instead of scalar consequents b_i, use granules B_i (interval/fuzzy/probabilistic):

$$B_i = [\underline{b}_i, \overline{b}_i] \quad \text{(interval case)}. \tag{14}$$

Then model output becomes a granular object, e.g., interval:

$$\hat{Y}(x) = \left[\frac{\sum_i w_i(x)\underline{b}_i}{\sum_i w_i(x)}, \frac{\sum_i w_i(x)\overline{b}_i}{\sum_i w_i(x)} \right]. \tag{15}$$

This converts "single value" prediction into "value + credibility span" prediction.

4.3 Principle of Justifiable Granularity

A granule should balance:

- **coverage** $\mathrm{cov}(G)$: how much data are represented,
- **specificity** $\mathrm{sp}(G)$: how informative/narrow the granule is.

A standard construction objective is:

$$G^* = \arg\max_G \; V(G), \qquad V(G) = \mathrm{cov}(G) \cdot \mathrm{sp}(G). \tag{16}$$

This is the key mathematical principle used to build interval/fuzzy granules in the 2025 TFS paper.

4.4 Probabilistic Granules via Gaussian Process

For GP regression:

$$y(x) \mid \mathcal{D} \sim \mathcal{N}(\mu(x), \sigma^2(x)). \tag{17}$$

A $(1-\alpha)$ confidence interval is:

$$\hat{Y}_\alpha(x) = \left[\mu(x) - z_{\alpha/2}\sigma(x),\ \mu(x) + z_{\alpha/2}\sigma(x)\right]. \tag{18}$$

The paper further maps probabilistic granules to interval/fuzzy granules through justifiable granularity criteria.

5 A Deeper Synthesis: What Each Family Optimizes

- **logic-net**: optimize predictive fit while distilling rule-regularized posterior structure.
- **semantic loss**: optimize probability mass over logically valid worlds.
- **DL2**: optimize satisfiability of declarative constraints in both training and querying.
- **granular TS/GC**: optimize not only fit, but also the informativeness of uncertainty representation.

So, the progression is:

$$\text{validity of output} \;\Rightarrow\; \text{validity + confidence geometry of output}. \tag{19}$$

6 A Research Blueprint

A mathematically strong formulation for my own work can be:

$$\min_{\theta,\eta} \underbrace{\mathcal{L}_{\text{task}}(\theta)}_{\text{fit}} + \lambda_1 \underbrace{\mathcal{L}_{\text{logic}}(\theta)}_{\text{rule satisfaction}} + \lambda_2 \underbrace{\mathcal{L}_{\text{gran}}(\theta,\eta)}_{\text{credible granules}} + \lambda_3 \underbrace{\Omega(\theta,\eta)}_{\text{regularization}} , \tag{20}$$

where η parameterizes interval/fuzzy/probabilistic output structure.

Recommended evaluation should report at least:

- prediction quality (Acc/F1/RMSE),
- rule satisfaction rate,
- granule quality (coverage, specificity, and calibration metrics).